

Design Before You Build — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Each line is one mark, tagged [1]. On the calculation questions award method marks for the right formula and the right substitution even when the arithmetic fails. On UI and UX questions a bare label never scores: the reason is the mark. The two mistakes this unit produces most are putting ideas in Define, and judging a fix by what can be seen on the screen. Every note names the specific error its question was built to catch. On the sketch questions, any colour, shading or real picture caps the mark: that is the polished-too-early trap.

Paper A — Medium

30 MARKS

- 1
 - (a) **Discover**, then **Define**, in the first diamond [1] [4]
 - (a) **Develop**, then **Deliver**, in the second diamond [1]
 - (b) **diverge, converge, diverge, converge**, left to right [1]
 - (c) **one clearly defined problem** (a problem statement) [1]

The usual slip is Define and Develop swapped, because both sound like making things. Award the two phase-name marks only if the order is right from left to right. A student who writes “the answer” or “a solution” for (c) has described the end of the second diamond; no mark.
- 2

Watch six students... = **Discover** [1] [4]

One sentence about late returns = **Define** [1]

Sketch eight different ways = **Develop** [1]

Release and track for a month = **Deliver** [1]

The trap pair is Define and Develop. A sentence that names the problem is Define; sketching eight options is Develop. If a student writes Develop for the one-sentence card, give nothing for either of those two rows: they have not separated the problem from the answer.
- 3
 - (a) **UI**: darker text changes how the screen looks (colour and contrast) [1] [3]
 - (b) **UX**: fewer taps changes how the task goes [1]
 - (c) **UX**: it is feedback, telling the user the tap worked [1]

Part (c) catches “I can see it, so it is UI”. A message is something you see, but its job is to say the tap worked, which is UX. Each mark needs the label and a reason; a bare label scores nothing.
- 4
 - (a) $(1 + 0.05) \div (0.10 + 0.05) = 1.05 \div 0.15 = 7.0$ [1] [4]
 - (b) $1.05 \div (0.25 + 0.05) = 1.05 \div 0.30 = 3.5$ [1]
 - (c) **Sample A**: 7.0 is at least 4.5; sample B at 3.5 is below it [1]
 - (d) **Yes**: 28 px is large text, and 3.5 is at least 3 [1]

Leaving out the + 0.05 on both parts gives $1 \div 0.10 = 10$ and $1 \div 0.25 = 4$, and 4 looks like “nearly passes”. Give no mark for (a) or (b) without the 0.05s, then mark (c) and (d) as follow-through on the student’s own figures.
- 5
 - (a) $(48 - 32) \div 2 = 8$ **px** each side [1] [3]
 - (b) $(48 - 28) \div 2 = 10$ **px** each side [1]
 - (c) **40 wide and 36 high**: it does **not** meet 48×48 [1]

Answers of 16 and 20 mean the extra was not halved: padding goes on two sides. In (c) 4 px on each side adds 8 to each measurement, not 4; a student who writes 36 and 32 has added 4 once.

- 6 (a) $X = (390 - 342) \div 2 = 24$ [1] [4]
 (b) Each step down is $96 + 24 = 120$ [1]
 (b) $Y = 88 + 2 \times 120 = 328$ [1]
 (c) Bottom of card 3 = $328 + 96 = 424$; $844 - 424 = 420$ px [1]
Three wrong methods, each visibly wrong against the diagram: $88 + 2 \times 96 = 280$ (forgot the gaps), $88 + 3 \times 120 = 448$ (three steps down, but card 3 is only two steps below card 1), and $844 - 328 = 516$ (measured from the top of card 3, not its bottom).
- 7 (a) First inconsistency, read from the picture (see note) [1] [4]
 (a) A second, **different** inconsistency [1]
 (b) Red alone fails for users who cannot tell red from other colours, in bright sunlight, or on a screen reader [1]
 (b) Fix: an **icon and a message** under the box saying what to fix [1]
Real differences in Figure 1: shape (Log in is a pill, Sign up is square, Guest is slightly rounded); colour (blue, green, grey); height (the three buttons are three different heights); capitals (CONTINUE AS GUEST). Two answers about the same property, such as blue and green, are one answer. "Make the red darker" is still colour alone: no mark for the fix.
- 8 A **navigation bar** across the top of the frame [1] [4]
 Exactly **one heading**, placed before the content [1]
 One **button** with a verb label (Renew, Search), not "Click here" or "Submit" [1]
 A **footer** as the last box on the page [1]
Any colour fill, shading, real picture or drawn-in artwork caps the question at 3: this is the over-polished-wireframe trap, and the question tells him so twice. Unlabelled boxes cannot be marked; a box has to say what it is.

Paper B — Medium

30 MARKS

- 1 (a) **No opening up and narrowing**: every phase is drawn the same, as one run, with no diamond shape [1] [4]
 (a) **No way back**: arrows only go forwards, but a test result can send the team to an earlier phase [1]
 (b) **Two diamonds** side by side, each opening up then narrowing, with the four phases named in order [1]
 (b) A **backward arrow** from a late phase to an earlier one [1]
The straight line is the commonest wrong picture of the double diamond. Credit (a) only for the two ideas, shape and loop-back; "it is too simple" or "it has no colours" earns nothing. Both parts (b) marks can be earned even if (a) was weak.
- 2 Sketch six different ways = **Develop** [1] [3]
 The "Commuters miss trains because..." sentence = **Define** [1]
 Group forty notes into three themes, then pick one = **Define** [1]
Rows 1 and 2 are the classic swap. The sentence names a problem; sketching six ways is a solution. Row 3 narrows towards a problem, which is why a student who thinks "Define is about ideas" gets it wrong too.
- 3 (a) **(ii)**, the blue Book button: it changes how the screen looks [1] [3]
 (b) (i) is UX: one screen instead of three changes how the task **goes** [1]
 (b) (iii) is UX: the spinner is **feedback**, telling the user the tap worked; it is seen, but its job is the experience [1]
The student in the question uses "you can see it" as the test, which is the wrong test: judge a change by what it is for. The spinner is the case that exposes it. Do not accept "(iii) is UI because it is a picture".

- 4 (a) $(0.35 + 0.05) \div (0.05 + 0.05) = 0.40 \div 0.10 = \mathbf{4.0}$ [1] [5]
 (b) $(1 + 0.05) \div (0.15 + 0.05) = 1.05 \div 0.20 = \mathbf{5.25}$ [1]
 (c) Button 1 **fails** and button 2 **passes** [1]
 (c) Both are compared with **4.5**, because 16 px regular is not large text [1]
 (d) 4.0 is below 4.5, so it fails: a minimum is a minimum, and text just under it is hard to read for people with weaker sight [1]
Leaving out the + 0.05 gives $0.35 \div 0.05 = 7$ for button 1, which would pass: the wrong method turns a fail into a pass, so check the working before the verdict. Mark (c) as follow-through if a ratio was wrong, but only with the 4.5 stated.
- 5 (a) $6 \times 44 = 264$ [1] [3]
 (a) $5 \text{ gaps} \times 12 = 60$, so the total is **324 px** (fits in 328) [1]
 (b) $7 \times 44 + 6 \times 12 = \mathbf{380 \text{ px}}$, which is more than 328: no [1]
Six gaps instead of five gives 336, more than 328, and wrongly says six buttons do not fit. n buttons have $n - 1$ gaps; a student who draws six buttons will see it.
- 6 (a) Row 5 is four rows below row 1 [1] [3]
 (a) $Y = 72 + 4 \times 56 = \mathbf{296}$ [1]
 (b) $(844 - 72) \div 56 = 772 \div 56 = 13.79$, so **13 whole rows** [1]
 $72 + 5 \times 56 = 352$ is the Y of row 6. Rounding 13.8 up to 14 counts a row that is cut off by the bottom of the frame; the question says whole rows.
- 7 (a) **Step 4, Passenger details** (score 1) [1] [4]
 (b) $(5 + 4 + 2 + 1 + 4 + 5) \div 6 = 21 \div 6 = \mathbf{3.5}$ [1]
 (c) A UX fix that changes the task, e.g. save family members so their details fill in automatically [1]
 (d) A detail *with the decision it changes*: slow data (light pages, save progress), or cracked screen and one finger (big targets, few fields) [1]
Yellow is the decoy: a detail belongs on a persona only if it changes a decision, and nothing depends on a favourite colour. A student who divides 21 by 5 steps and gets 4.2 has miscounted: the table has six. A UI fix in (c), such as a bigger font, does not score.
- 8 **Same navigation, same place**, on all three pages [1] [5]
 Exactly **one heading** per page, before the content [1]
Home has a button after its heading [1]
Contact has a form with two fields and a button, and no other page has a form [1]
 A **footer** that is the last box on every page [1]
These are the five rules of the page's wireframe studio; mark them rule by rule from the frames. A nav bar that moves between pages fails mark 1 even if it is at the top on two of them. Any colour, shading or real picture caps the question at 4: that is the polished-too-early trap.

Paper C — Hard

30 MARKS

- 1 (a) **Discover**, then **Define**, in the first diamond [1] [4]
 (a) **Develop**, then **Deliver**, in the second diamond [1]
 (b) **diverge, converge, diverge, converge**, left to right [1]
 (c) An arrow from the end back to **Discover**: the cause is new evidence nobody had gathered [1]
"Define" alone earns no mark in (c): a team cannot rewrite a problem it has not yet researched, and the question says the cause was never researched. Accept Define only if the student draws the arrow to Discover first. A straight arrow forward is the straight-line trap in another form.

- 2 (a) **Develop**, and **diverging** (many ideas for a solution) [1] [3]
 (b) Define should have ended with **one clear problem statement** [1]
 (b) Without it there is no way to choose among the thirty ideas, or they may all solve the wrong problem [1]
The trap is that both phases start with D and both feel like making things. Thirty ideas open the options up, so it cannot be Define, which narrows. "They are in Define because they are choosing" gets nothing for (a). Reasons about time or tidiness in (b) do not score.
- 3 (a) **UX**: it shows how far and how much is left, which changes whether people finish [1] [3]
 (b) **UI**: text size is typography, a change to how it looks [1]
 (c) **UX**: one tap instead of a whole order changes how the task goes [1]
A progress bar looks like decoration, so students write UI. Each mark needs the label and a reason that names the task; "UX because it helps people" is too vague. Part (b) is the control: a bigger font is genuinely UI, so a student who writes UX for everything has not learnt the test.
- 4 (a) $(0.45 + 0.05) \div (0.05 + 0.05) = 0.50 \div 0.10 = 5.0$ [1] [5]
 (a) $(0.30 + 0.05) \div (0.05 + 0.05) = 0.35 \div 0.10 = 3.5$ [1]
 (a) $(0.85 + 0.05) \div (0.10 + 0.05) = 0.90 \div 0.15 = 6.0$ [1]
 (b) **Sample ii**: 3.5 is below 4.5 but at least 3 [1]
 (c) Looking clear is not a measurement: screens, sunlight and eyes differ, so the ratio decides [1]
The lighter colour goes on top. In sample i the text is the lighter colour; a student who always puts the background on top gets $0.10 \div 0.50 = 0.2$, an impossible ratio (a ratio is never below 1), so point that out rather than only marking it wrong. Samples i and iii pass; only ii sits between 3 and 4.5.
- 5 (a) Usable width = $360 - 2 \times 16 = 328$ [1] [5]
 (a) $3 \text{ gutters} \times 8 = 24$; $(328 - 24) \div 4 = 76 \text{ px}$ [1]
 (b) $2 \times 76 + 8 = 160 \text{ px}$ [1]
 (c) Column 3 starts after two columns and two gutters: $16 + 2 \times (76 + 8)$ [1]
 (c) = **184 px** [1]
Three visible slips: $328 \div 4 = 82$ (ignores the gutters), $2 \times 76 = 152$ (a two-column card spans one gutter), and $16 + 2 \times 76 = 168$ (forgets the two gutters before column 3). Check: two 2-column cards, $160 + 8 + 160 = 328$, fill the usable width exactly.
- 6 **Home**: no button after the heading (Home must have a call to action) [1] [4]
About: no navigation (same navigation at the top of every page) [1]
About: two headings (exactly one per page) [1]
Contact: a paragraph after the footer, so the footer is not last [1]
Each flag needs the page and the rule. Home's navigation and heading are fine, and Contact is where the form belongs: a student who flags the form or the image box has flagged correct things, and earns nothing for them. About's two faults are separate marks.
- 7 (a) **Polishing the wireframe too early** [1] [3]
 (b) Colour and photographs pulled attention to decoration, so people argued about the shade and missed the missing structure [1]
 (c) A **low-fidelity wireframe**: grey boxes and labels, so feedback is about layout and flow [1]
The effort is not the mistake; the timing and the fidelity are. "He should have worked less" or "he should have asked first" earns nothing. The missing route to the menu is the proof for (b): a plain wireframe would have made the gap the only thing to look at.

- 8** **Who** is affected (the parents of the bus families, not “everyone”) and **what goes wrong** in plain words **[1]** [3]
 Uses a figure from the research **correctly**: 27 of 36 (75%), 18 minutes, or 10 to 25 minutes late **[1]**
 Says **why it matters**, and contains **no solution** **[1]**
Model: “Parents of the bus families, 27 of 36 of them, wait about 18 minutes at the gate because the bus is 10 to 25 minutes late, so they lose working time and children wait outside.” Any solution word (app, tracker, SMS, GPS, “should build”) loses the third mark whatever else the sentence says. $27 \div 36 = 0.75$, so 75%; 27% or 36% means the two figures were swapped.

Paper D — Hard

30 MARKS

- 1** (a) A box for **each of the four pages**, Home at the top, joined to the others **[1]** [4]
 (a) The links or a note showing the navigation connects **every page to every other page** **[1]**
 (b) Before: 3 pages $\times$ 2 other pages = **6 links** **[1]**
 (b) After: 4 pages $\times$ 3 other pages = **12 links** **[1]**
 $4 \times 4 = 16$ counts every page linking to itself, which nobody needs. Notice that one new page adds 6 links, not one: every old page needs a link to it, and it needs links to all three. A sitemap drawn as a tree with Home above three boxes earns the first mark; the second needs the all-to-all idea shown somewhere.
- 2** (a) **Define**: grouping the notes to find the pattern narrows **[1]** [3]
 (b) **Develop**: drawing and testing different layouts keeps the options open **[1]**
 (c) **Define**: the problem statement has to be rewritten **[1]**
Part (c) is the loop-back. The team has the facts and has a design that works; what is wrong is the problem statement, so they go back to Define. Discover is wrong because they already have the evidence, and Develop is wrong because they have a good design for the wrong problem.
- 3** (a) $(1 + 0.05) \div (0.25 + 0.05) = 1.05 \div 0.30 = 3.5$ **[1]** [3]
 (b) **16 px regular fails**: 3.5 is below 4.5 and 16 px is not large text **[1]**
 (b) **20 px bold and 24 px regular both pass**: bold from 18.66 px, or any text from 24 px, counts as large, and 3.5 is at least 3 **[1]**
The hard paper gives no thresholds, so this tests recall. Students who remember 24 px but not 18.66 px bold will say 20 px bold fails: award the third mark only if both passes are named.
- 4** (a) $844 - 72 = Y = 772$ **[1]** [5]
 (b) Rows fill $Y = 96$ to $Y = 772$: 676 px; $676 \div 64 = 10.56$, so **10 whole rows** **[1]**
 (b) Left over: $676 - 10 \times 64 = 36$ px **[1]**
 (c) $676 \div 72 = 9.39$, so **9 whole rows** now fit **[1]**
 (c) **All of the instances** change, because editing the main component updates every instance **[1]**
Dividing $(844 - 96) \div 64$ gives 11 rows: the tab bar was forgotten, and the last row would sit underneath it. Rounding 9.39 up to 10 keeps a row that is cut off. In (c) a student who says only one instance changes has copied rectangles; that is the exact mistake components exist to prevent.

- 5** (a) Before: $16 \div 5 = 3.2$ [1] [5]
 (a) Fix A: $20 \div 5 = 4.0$; Fix B: $18 \div 5 = 3.6$ [1]
 (b) **Step 3, Sign in**, scoring 1; Fix A **leaves it at 1** [1]
 (c) Fix B lifts the lowest point from 1 to 3 [1]
 (c) An average can rise while the one step that makes people quit stays broken: anyone who stalls at sign-in never reaches the steps Fix A improved [1]
The question is built so the higher average (4.0 against 3.6) points at the worse fix. The lowest point is where you fix first; the average hides it. A student who picks A because 4.0 is bigger than 3.6 has used the average and should see the sign-in score of 1 before the explanation.
- 6** The **page list and file names** (index.html, about.html, contact.html) [1] [4]
 Where **each navigation link** goes [1]
 The **frame size** the pages were designed at, e.g. 390×844 [1]
 Which text is **placeholder** and which is final, **or** what each image should show, **or** the order of blocks on each page [1]
Accept any four different items from: page list and file names; where each link goes; the frame size; the order of blocks on each page; which text is placeholder; what each image should show. Not accepted: colours and fonts (a low-fidelity wireframe has none, deliberately) or "how to code it", which is Kabir's job.
- 7** (i) The mistake: **UI and UX are swapped** [1] [6]
 (i) Correct: **UI is what you see and touch; UX is how the whole thing works and feels** [1]
 (i) Applied: changing a button's colour changes how the screen looks, so it is a **UI** fix [1]
 (ii) The mistake: **colour is the only signal** [1]
 (ii) Correct: show it in red **and** with an icon and a message saying what to fix [1]
 (ii) A reason: some users cannot tell red from other colours (about 1 in 12 men), screens wash out in sunlight, screen readers do not see colour [1]
Both statements are the page's named traps, so the diagnosis has to be specific. In (i) the conclusion is also wrong, and a student who corrects the definition but keeps "UX fix" has not finished: give the third mark only for applying the corrected idea. In (ii) "use a brighter red" is still colour alone, so no mark for the correction. Marking rubric for each: names the error [1], writes a correct version [1], gives a reason or test that would convince Rohan [1].